ML15 A complete regression experiment

Week 5, Baselines and a line | Session 15 of 24 | Machine Learning | New library: sklearn

Today's question. Does the rent model beat a plain guess on homes it has never seen?

Objective. By the end: you can split data, fit a model and a baseline on the training part, and compare their errors on held-out homes.

> **WHERE THIS IS USED**
> A model can look perfect on the homes it learned from and still fail on a new one. Teams that build tools for lending, fares or weather test on cases kept aside. Would you trust a model that was only tested on its own homework?

Before this session: ML14 Predicting a number with a line. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
train and test split	Divide the rows into homes to learn from and homes to try the model on.	One pile to practise on and one pile kept for trying it out.
held-out records	The rows kept back and never used to fit.	A pile of cards kept in a drawer until the end.
test_size	The share of rows kept back for testing.	One quarter of the cards go in the drawer.
random_state	A number that makes the shuffle repeatable.	Giving everyone the same shuffled deck.
baseline model	A model that guesses the training mean every time.	The average guess.
mean absolute error	The average size of the misses, in the target's own unit.	The average distance of your darts from the bullseye.

Block A. Split the homes

Keep a quarter of the 120 homes back. The model must never see them while it learns.

Predict. Predict how many homes go to training and how many to testing. Write your guess on paper first.

Your guess, written before you run it

```
import pandas as pd
from sklearn.model_selection import train_test_split
homes = pd.read_csv("synthetic_homes.csv")
X = homes[["area_m2", "rooms"]]
y = homes["rent_ugx"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25, random_state=7)
print(len(X_train), len(X_test))
```

Run it, then compare. The cell prints 90 and 30. The four names on the left receive training inputs, test inputs, training targets and test targets, in that order. random_state=7 fixes the shuffle so everyone gets the same split.

Investigate.

1. Why do we shuffle before splitting?
2. Which of the four names would hold the answers you must not peek at?

Modify. Change test_size to 0.2 and run the split again.

Block B. Baseline against model

Fit a mean-guess baseline and a line on the 90 training homes. Measure both on the 30 held-out homes.

Predict. Predict whether the model's average miss will be smaller than the baseline's, and by roughly how much. Write your guess on paper first.

Your guess, written before you run it

```
from sklearn.dummy import DummyRegressor
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error
baseline = DummyRegressor(strategy="mean").fit(X_train, y_train)
model = LinearRegression().fit(X_train, y_train)
baseline_mae = mean_absolute_error(y_test, baseline.predict(X_test))
model_mae = mean_absolute_error(y_test, model.predict(X_test))
print(round(baseline_mae), round(model_mae))
```

Run it, then compare. The cell prints 110169 and 11889. The baseline misses by about UGX 110169 and the model by about UGX 11889, so on these 30 held-out homes the model is far better. Mean absolute error averages the misses and uses the unit of the target, shillings. The model wins by this much because the homes were generated from a formula, which makes them easy to learn. Real data would not be so kind.

Investigate.

1. What does a mean absolute error of 11889 mean in words?
2. Why is this result no evidence about real rents?

Modify. Fit a model using only area_m2 and print its error.

Make

Run the experiment, record the two errors, and say whether the model beats the baseline on the held-out homes. Keep the split as it is. Do not keep changing random_state to get a better score.

Save your work as ML15. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
KeyError: "['rooms'] not in index"	The column name is spelled differently from the header.	Print homes.columns and copy the exact name.
The two errors are the same	The model and baseline were scored on the same predictions, or the wrong table was used.	Check that each uses its own predict call on X_test.

Exit ticket

On held-out homes the model's average miss is about a tenth of the baseline's, and this holds only for these invented homes.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.